Mingxi Yan

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Summary

AI engineer and researcher specializing in Agents, LLM applications, evaluation, and retrieval. Experience spans product workflows and academic research, including hands-on software development, agentic training, and model fine-tuning.

Education

Carnegie Mellon University **Aug 2026 - Aug 2027**
Master of Science, Language Technologies (Artificial Intelligence) *Pittsburgh, PA*

Carnegie Mellon University **Aug 2022 - May 2026**
Bachelor of Science, Computer Science and Information Systems; Minor in Logic and Computation *Pittsburgh, PA*
- **GPA:** 3.86 / 4.0

Skills

- **Programming**: Python, SQL, C, JavaScript, TypeScript, Data Structures, Database Design, Distributed Systems, Web Application
- **Machine Learning**: PyTorch, vLLM, Hugging Face, TRL, PEFT/LoRA, SFT, DPO, Transformers, Multimodal ML
- **Agents**: Agentic Search, Agent Memory, Agenti RAG, Agentic RL, tool use, LLM evaluation, Skills, MCP, Agent Safety
- **Data**: PostgreSQL, SQLite, Docker, AWS, Search, Feature Engineering, data pipelines, information retrieval, DataOps

Industry Experience

TikTok **Jun 2026 - Aug 2026**
Data Engineer Intern *San Jose*
- Built a benchmark using on-call Q&A pairs to evaluate agent results. Integrated as an internal published skill.
- Designed a graph-based enterprise retrieval skill over Hive tables, datasets, dashboards, documents, and millions of metadata fields and values using query decomposition, contextual rewriting, graph expansion, and RRF. Improved accuracy by 23%.
- Built a livestream revenue diagnosis copilot, using feature engineering, LightGBM percentile prediction, comparison cohorts, and business rules to detect revenue downside, identify abnormal metrics, retrieve relevant business documentation, and generate grounded recommendations. Validated with operations stakeholders and integrated into their workflow.

Enterview (AI Startup) **Jun 2023 - Nov 2024**
Full Stack Engineer & Tech Lead *Pittsburgh, PA*
- Led a four-engineer team building an AI interview-preparation platform while partnering on technical product decisions.
- Designed persistent data models to store customized user information and backend APIs for personalized interview coaching, integrating an LLM and a speech recognition model to evaluate spoken responses and generate targeted feedback.

Sapling Lab, Carnegie Mellon University **May 2026 - Present**
Research Assistant *Pittsburgh, PA*
- Led NLP research for an NSF and OpenAI-funded interdisciplinary AI safety project studying how users form and update Theory-of-Mind attributions toward AI during long-term human–AI interaction. Designed, conducted, and analyzed data from a human study.
- Developed a multi-turn user simulation to study user traits, assistant social behavior, memory, and scenarios; created a human-calibrated LLM judge; and distilled a cheap, fast decision model for online evaluation.

Chimps Lab, Carnegie Mellon University **May 2024 - May 2026**
Research Assistant *Pittsburgh, PA*
- Co-authored PersonaTeaming (UIST 2026) in collaboration with Apple Research, designing the persona-generation algorithm for adaptive LLM red-teaming. Evaluated 11 experimental conditions on 6 different LLM targets. Improved mean attack success rate by 46% and attack diversity by 7% over RainbowPlus (baseline).
- Built scalable image-generation bias-auditing infrastructure with persistent history, caching, batching, supporting 100 images per prompt. Designed interactive visualizations for researchers and participants to cooperate.

ModelBest, THUNLP Lab, Tsinghua University **May 2023 - Aug 2023**
Research Intern *Beijing*
- Converted REST APIs into LLM-callable tools with searchable metadata and instructions; built tests for tool invocation.
- Applied embedding-based clustering to categorize APIs and improve tool retrieval.

Selected Course Project

Solver-Guided Fine-Tuning for Poker Explanations **Jan 2025 - May 2025**
Carnegie Mellon University *Pittsburgh*
- Designed and implemented a solver-guided post-training pipeline for Llama 3 8B, constructing 3,000 SFT examples and 2,534 DPO pairs from PokerBench solver labels and verifier-ranked Llama 3.3 70B explanations.
- Implemented LoRA SFT/DPO and held-out evaluation, improving exact solver-action match from 18.0% to 56.2% and verifier score from 0.426 to 0.645 with SFT; identified an explanation-quality versus action-accuracy tradeoff under DPO.